

Laboratory on Linux basics (G exercises)

Install Linux in case you do not already have it installed (Tip, Ubuntu:

<http://www.ubuntu.com/desktop/get-ubuntu/download>).

You can either make a clean installation or run it through virtualization by first installing VirtualBox from: **<http://www.virtualbox.org/>**.

Note: Only if you intend to use your own computer, Linux is already installed on the laboratory computers.

Some general information about Linux can be found here:

<http://www.linux.org/lessons/beginner/toc.html>

<http://www.linux-tutorial.info/modules.php?name=MContent&pageid=224>

<http://www.freeos.com/guides/lsst/>

Or simply search the web there is an abundance of information about Linux available.

Exercises

All answers are to be presented in your written report document.

Network configuration, Routing table and Virtual Interfaces

1) Introduction

If you have a network that ranges from 192.168.1.0 to 192.168.1.255 explain why individual devices in the network can only be assigned IP addresses in the range of 192.168.1.1 to 192.168.1.254. **(write down the answers in your written report).**

2) Find IP & MAC

Find out about network and hardware information for the computer you are currently using. **(Write down the IP and MAC address of your computer in your written report).**

3) Routing Table basics

Now, enter the command: **"\$ netstat -r"** to print your computers routing table. Explain (very briefly) the different columns: **Destination, Gateway, Genmask, Flags, MSS, Window, irtt and Iface**, **(write the answer in your written report in table form).**

4) Virtual Interfaces

Linux offers the possibility to set up interfaces according to your networking needs. For instance, if needed, you can configure an interface for multiple IP addresses by creating new virtual interfaces with another IP address.

a) Create a new virtual interface with following IP address, 192.168.2.32 and netmask 255.255.255.0 then check to see if the interface was created successfully? **(save a print screen of your interface table to display in your written report document).**

b) Now, you need to set up a route for this interface so that your computer can see it. Otherwise, everyone else on the network will be able to reach the new interface except you. Issue the needed command, then issue the "\$ netstat -r" command and check if the route to your added interface is visible **(save a print screen of your routing table to display in your written report document together with the command(s) you used to set up the interface).**

c) Next remove the route for this interface, **(write down the command(s) in your written report).**

d) Then remove the interface completely, **(write down the command(s) in your written report).**

5) Add a New Network

Add a new network. Obviously, first you need to be aware of your network address, netmask, gateway and what ethernet device you wish to set up for it. However, to keep things simple but still get a feel for the principle, in our case we will use the values for the primary network you belong to.

a) Enter the command needed to add another network with the same values as your primary network meaning: **(yourPrimaryNetworkAddress)** and **(netmaskForYourNetwork)**, **(write down the command(s) in your written report).**

b) Assign the default gateway for your newly added network (tip the same default gateway as you primary network), **(yourDefaultGatewayAddress)**

c) Look for your newly added network in your routing table by issuing the "\$ netstat -r" command. You should now have a double setup of your primary network in the table **(save a print screen of your routing table to display in your written report document).**

d) Now, remove your changes meaning the double routing table setup for your primary network. First issue the command needed to delete your newly added route then issue the command to delete your newly added default gateway. **(write down the command(s) in your written report)**

6) Multinetwork Scenario Configuration

You should now set up a working routing table for a multi-network scenario. Assume that you have two network cards available connected to two different LANs. The destination of the first network is, 10.0.2.0 with netmask 255.0.0.0 and the second, 192.168.1.0 with netmask 255.255.255.0. Furthermore, a firewall is assumed to exist between the two networks, where network card eth0 is attached to the 10.0.2.0 network and eth1 is attached to the 192.168.1.0 network. To forward packets on the Internet the firewall needs to route packets from the 10.0.2.0 network through the 192.168.1.0 network. The firewall system must be set up with two IP addresses, 10.0.2.1 on eth0 and 192.168.1.25 on eth1. The gateway to the Internet on the 192.168.1.0 network should be 192.168.1.1.

Provide the necessary commands to route on the firewall/router system:

- a)** Assign the firewall IP addresses to eth1 and eth2.
- b)** Add the routes for the networks, i.e., 192.168.1.0 on eth1 and 10.0.2.0 on eth0
- c)** Assign the Internet gateway (meaning: 192.168.1.1) as the default gateway.
(write down the command(s) in your written report)

d) Enter the necessary command(s) in order for packets belonging to computers in the 10.0.2.0 network to be routed to the 192.168.1.0 network and the Internet. In other words this should tell each computer on the 10.0.2.0, which the default gateway is, i.e., your firewall/router. You do not need to worry about the route back configuration it is enough to assign the proper default gateway for the 10.0.2.0 network.
(write down the command(s) in your written report)